precise flight path management, refueling coordination, and mission execution without continuous human intervention.

Technological innovations and capabilities

1. *Autonomous carrier operations*: The X-47B was the first UAV to demonstrate fully autonomous catapult-assisted takeoff and arrested landing (CATOBAR) from an aircraft carrier deck. Using advanced algorithms and navigation sensors, it successfully launched and recovered aboard the USS George H.W. Bush (CVN-77) in May 2013, a landmark achievement in naval aviation history.
2. *Aerial refueling autonomy*: In 2015, the X-47B became the first UAV to perform an autonomous aerial refueling using the US Navy's probeand-drogue system, extending its operational range and endurance, key capabilities for future long-duration missions.
3. *Stealth and survivability*: The UCAV's radar-absorbent structure, shielded exhausts, and internal weapon storage ensure low detectability in high-threat environments, supporting deep-strike and reconnaissance operations in anti-access/area-denial (A2/AD) regions.
4. *Network-centric integration*: The X-47B was designed to operate within a networked battle environment, exchanging data with other naval platforms, manned aircraft, and command centers in real time through secure datalink systems.
5. *Autonomous decision-making systems*: Equipped with AI-driven mission management software, the X-47B can make in-flight adjustments to optimize fuel efficiency, route planning, and threat avoidance while maintaining communication with command elements for strategic supervision.

Key specifications: